THE ABSOLUTELY WEAKLY NORMAL STRUCTURE SHEAF IS THE SHEAFIFICATION OF THE STRUCTURE SHEAF FOR THE H AND PH TOPOLOGIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EVU (sources/stacks-flat/pinned-original-source.tex line 12077).

Allowed standard prerequisites (editorial).

- The absolutely weakly normal structure sheaf is a ph sheaf, with a canonical map from the structure sheaf. Same chapter, Lemma labelled lemma-weak-normalization-ph-sheaf (source line 11991), invoked at proof lines 12089-12091. Supplies that the rule is already a sheaf for ph, hence for h, and the canonical map of presheaves whose sheafification property remains to be shown.
- The absolutely weakly normal quotient of a ring and its elements. Morphisms, Lemma labelled morphisms-lemma-seminormalization-ring (Stacks tag 0EUR), invoked at proof lines 12107-12108. Supplies that an element of F(Spec(A)) lies in A^{awn} and admits a finitely presented model.
- Realising an element as the image of a finitely presented universal homeomorphism. Morphisms, Lemma labelled morphisms-lemma-universal-homeo-limit (Stacks tag 0EUJ), invoked at proof lines 12109-12113. Supplies the ring map A -> B of finite presentation with Spec(B) -> Spec(A) a universal homeomorphism over which f comes from an element of B, giving the required h covering.
- The big and small h sites. Same chapter, Definition labelled definition-big-small-h (source line 10504), invoked in the statement. Fixes the site on which the sheafification statement is made.

Source reference. Original definition 0EU1: sources/stacks-flat/pinned-original-source.tex, source lines 10504–10504. Same-chapter pointer into the author environment definition at source line 10504; invoked at proof line None of the counted

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proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EVT: sources/stacks-flat/pinned-original-source.tex, source lines 11991–11991. Same-chapter pointer into the author environment lemma at source line 11991; invoked at proof line 12089 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EUR: sources/stacks-chow/tags.txt, source lines 18561–18561. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12108 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EUJ: sources/stacks-chow/tags.txt, source lines 18554–18554. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12109 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let S be a scheme. Choose a site $(Sch/S)_h$ as in Definition 0EU1. The rule*

$$X \longmapsto \Gamma(X^{awn}, \mathcal{O}_{X^{awn}})$$

is the sheafification of the "structure sheaf" $\mathcal{O}$ on $(Sch/S)_h$. Similarly for the ph topology.

Proof. In Lemma 0EVT we have seen that the rule $\mathcal{F}$ of the lemma defines a sheaf in the ph topology and hence a fortiori a sheaf for the h topology. Clearly, there is a canonical map of presheaves of rings $\mathcal{O} \to \mathcal{F}$. To finish the proof, it suffices to show

(1) if $f \in \mathcal{O}(X)$ maps to zero in $\mathcal{F}(X)$, then there is a h covering $\{X_i \to X\}$ such that $f|_{X_i} = 0$, and
(2) given $f \in \mathcal{F}(X)$ there is a h covering $\{X_i \to X\}$ such that $f|_{X_i}$ is the image of $f_i \in \mathcal{O}(X_i)$.

Let f be as in (1). Then $f|_{X^{awn}} = 0$. This means that f is locally nilpotent. Thus if $X' \subset X$ is the closed subscheme cut out by f, then $X' \to X$ is a surjective closed immersion of finite presentation. Hence $\{X' \to X\}$ is the desired h covering. Let f be as in (2). After replacing X by the members of an affine open covering we may assume $X = \mathrm{Spec}(A)$ is affine. Then $f \in A^{awn}$, see Morphisms, Lemma 0EUR. By Morphisms, Lemma 0EUJ we can find a ring map $A \to B$ of finite presentation such that $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is a universal homeomorphism and such that f is the image of an element $b \in B$ under the canonical map $B \to A^{awn}$. Then $\{\mathrm{Spec}(B) \to \mathrm{Spec}(A)\}$ is an h covering and we conclude. The statement about the ph topology follows in the same manner (or it can be deduced from the statement for the h topology). □